

Multilevel Determinants of Player Market Value in Elite European Football: The Role of Performance, Club Prestige, and Player Characteristics

Sagnik Chakrava
and
Jianing Zou
and
Kevin Linares
and
Nimit Shrivastava
University of Maryland

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Abstract

This study investigates the determinants of player market values in elite European football using a three-level cross-classified mixed-effects model. We analyzed 7,023 season-year observations of 2,215 players across 21 elite clubs from 2013 to 2024. The primary research question examines how individual performance metrics, player characteristics, and club-level prestige jointly predict market valuations. Results demonstrate that club prestige, measured by the UEFA 10-year coefficient, explains 47.9% of between-club variance in player values ($p < 0.001$), independent of individual performance. Additionally, each goal scored increases market value by 3.75% ($p < 0.001$), while each assist adds 4.95% ($p < 0.001$). Positional effects reveal that attackers command approximately 71.1% premiums over goalkeepers ($p < 0.001$). The full model achieves a conditional R^2 of 0.793. These findings provide evidence for a substantial “halo effect,” whereby players at elite clubs command valuation premiums independent of individual performance.

Keywords: football economics, multilevel modeling, player valuation, club prestige, UEFA coefficient, sports analytics

1 Introduction

The global football transfer market has grown exponentially, with annual expenditures exceeding €7 billion. Consequently, understanding the determinants of player market values is critical for clubs seeking competitive advantages. While traditional scouting emphasized subjective assessments, the “Moneyball” revolution demonstrated that data-driven approaches could identify undervalued players by focusing on objective metrics (Herm et al., 2014). Crowd-sourced platforms like Transfermarkt provide real-time valuations that correlate strongly ($r > 0.90$) with actual transfer fees (Herm et al., 2014).

However, player market values likely reflect contextual factors such as club reputation and competitive success, in addition to individual performance. Previous research suggests that 15-20% of salary variation is attributable to team-level factors (Franck & Nüesch, 2012). This raises a critical question: to what extent do club-level characteristics create a “halo effect” on player valuations? This study addresses three objectives:

1. **Quantify individual performance effects:** Determine the marginal impact of measurable metrics (goals, assists) on market values.
2. **Assess club prestige effects:** Estimate the influence of club-level prestige (UEFA coefficients) on valuations beyond individual characteristics.
3. **Model multilevel structure:** Account for the cross-classified nature of football data, where repeated measurements are nested within players, and players transfer between clubs.

2 Theoretical Framework

We integrate human capital theory with signaling theory. Human capital theory posits that market value reflects an individual’s productivity potential (goals, assists). Signaling theory suggests that club affiliation serves as a quality signal; playing for elite clubs indicates higher quality because these clubs maintain higher selection standards.

Herm et al. (2014) established the validity of crowd-sourced market valuations, identifying performance, age, and club prestige as key attributes. While performance metrics consistently emerge as significant predictors, the relative importance of different dimensions remains contested. Furthermore, prior research rarely employs multilevel models capable of attributing variance to individual versus team-level factors.

Franck and Nüesch (2012) demonstrated that team-level factors like brand value significantly influence compensation. However, most research treats club affiliation as a fixed effect or omits it, potentially biasing estimates. Additionally, few studies account for player mobility, which violates strict nesting assumptions. To address this, we utilize the UEFA coefficient as an objective measure of club prestige based on historical performance. We propose the following hypotheses:

- **H1:** Player performance metrics will exhibit significant positive associations with market value.
- **H2:** Club-level UEFA coefficients will predict player market values beyond individual characteristics.
- **H3:** Player age will demonstrate a negative relationship with market value.

- **H4:** Attacking players will command significant premiums over defensive players.

3 Methods

Data Source and Sample Data were obtained from Transfermarkt and UEFA official rankings. The sample consists of 7,023 season-year observations representing 2,215 unique players across 21 elite European football clubs from 2013 to 2024. Clubs were selected based on consistent presence in top UEFA rankings.

Variables

- **Outcome:** *Market Value* (2024 USD), log-transformed to address right skewness.
- **Level 1 Predictors (Time-Varying):** *Season Year* (2013-2024), *Goals*, *Assists*, and *Yellow Cards*.
- **Level 2 Predictors (Player-Invariant):** *Position* (Goalkeeper, Defender, Midfield, Attack), *Age* (grand-mean centered at 26.5), *Height* (grand-mean centered at 182.4 cm), *Birth Region*, and *Citizenship of Club Country*.
- **Level 3 Predictor (Club-Level):** *UEFA Coefficient* (grand-mean centered at 194.36).

Data Structure The data exhibit a cross-classified structure. Measurements (Level 1) are nested within players (Level 2). However, players are not strictly nested within clubs because 456 players (20.6%) transferred clubs during the observation window. We model player and club as crossed random effects: (1 player_id)+(1 club_id).